INCH-POUND

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MILITARY SPECIFICATION

QUADRANT, FIRE CONTROL: M18

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of fire control quadrant which provides a measure of elevation angles. It is equipped with numerical counters which indicate elevation settings and corrections. The level vial and counters are radioactively illuminated. (See 6.1)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-P-14232	-	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1290

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on date of the solicitation.

DRAWINGS

U.S. Army Armament Research, Development and Engineering Center (ARDEC)

F8565556	-	Pressure Tester
A8658940	-	Gunners Quadrant, Calibrated (Inspection Aid)
F11729525	-	Quadrant, Fire Control: M18
F11729722	-	Fixture, Elevation
F11747194	-	Final Inspection Fixture for Quadrant, Fire Control: M17
F11747195	-	Vibration and/or Shock Fixture for Quadrant, Fire Control: M17 and/or M18
B11747939	-	Adapter, Torque
B11747940	-	Adapter, Torque
F11747969	-	Gage, Interchangeability, Maximum

PACKAGING DATA SHEET

11729525 - Packaging of Quadrant, Fire Control: M17

(Copies of drawings and packaging data sheets required by suppliers in connection with specific procurment functions should be obtained from the procuring activity or as directed by the contracting officer.)

DEPARTMENT OF DEFENSE, DEFENSE SUPPLY AGENCY

DSAM 4145.8 - Radioactive Commodities in the DoD Supply System.

(Application for copies of the DSAM should be addressed to the Superintendent of documents, Government Printing Office, Washington, DC 20402.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Hazardous material. This quadrant contains tritium, a radioactive material. A Nuclear Regulatory Commission License is required for the manufacture of this item, and DSAM 4145.8 for procurement and possession of radioactive material.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.2 and 6.2).

3.3 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.4 Fabrication. The quadrant shall be manufactured in accordance with Drawing F11729525.

3.5 General specifications. The contractor shall be responsible for the compliance with the requirements of specification MIL-F-13926.

3.6 Environmental.

3.6.1 Storage temperature. The quadrant shall show no evidence of physical failure when thermally stabilized at ambient temperatures of +160°F (+71°C) ± 5°F (± 3°C) and -50°F (-46°C) ± 5°F (± 3°C) for a period of 4 hours ± 15 minutes at each temperature.

3.6.2 Operating temperature. The quadrant shall meet the applicable requirements of 3.12.1 while exposed and thermally stabilized for 2 hours ± 15 minutes at an ambient temperature of +150°F (+65°C) ± 5°F (± 3°C) and exposed and thermally stabilized for 4 hours ± 15 minutes at an ambient temperature of -50°F (-46°C) ± 5°F (± 3°C). Upon return to standard ambient temperature, +60°F to +90°F (+16°C to +32°C), from each operating temperature, the mount shall meet the requirements of 3.7 thru 3.15.

3.6.3 Shock. The quadrant shall withstand a total of 18 shock impulses, 3 in each direction of 3 mutually perpendicular axes. Each shock impulse shall be a half-sine wave with a time duration of

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6 + 1 milliseconds. The peak amplitude of each shock impulse shall be 75 g's. Subsequent to shock, the telescope shall show no evidence of physical damage and shall meet the requirements of 3.7 through 3.15.

3.6.4 Vibration "A". Where required, the quadrant shall withstand a total of 270 minutes + 5 minutes of sweep-cycle vibration. The vibration shall be applied for 90 minutes + 2 minutes along each of the three mutually perpendicular major axes. A complete sweep-cycle shall consist of vibration from origin (5 hz at 1 inch double amplitude) to mid-point (5 g's + 0.5 g's at 500 hz) to origin, and shall have a duration of 15 minutes + 1 minute. Double amplitude shall be constant at 1 inch between 5 hz and 10 hz, and varied with frequency to maintain a constant 5 g's + 0.5 g's acceleration between 10 hz and 500 hz. Upon completion of vibration, the quadrant shall exhibit no evidence of damage and shall meet the requirements of 3.7 through 3.15.

3.6.5 Vibration "B". The quadrant shall be vibrated in a vertical plane at a constant frequency of 30 + 5 cycles per second with an amplitude of 1/16 inch (1/8 + 1/64 inch total excursion) for a period of five minutes + 15 seconds. Subsequent to vibration, there shall be no evidence of physical failure and the quadrant shall meet the requirements of 3.7 through 3.15.

3.7 Sealing and purging.

3.7.1 Sealing. The quadrant shall show no evidence of leakage when subjected to an internal pressure of 5 + 0.10 pound per square inch gage (psig) for a minimum of one hour.

3.7.2 Purging. The interior of the quadrant shall be purged with dry nitrogen supplied at a pressure of 8 + 1/4 psig until the dew point of the emergent gas is no higher than -25°F. Purging apparatus shall be removed, and subsequent to the seating of the pressure relief valve, the quadrant shall be pressured to 1.0 + 0.1 psig for all testing and delivery.

3.8 Cleanliness.

3.8.1 Defects. There shall be no evidence of dirt, moisture, finger prints, condensates, or fractures on glass windows or counters. Internal inspection shall be made by viewing through the windows.

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3.9 Orientation. The requirements of 3.10.1 through 3.11.4 shall be met with:

a. The flat mounting surface of the quadrant in a vertical plane.

b. The locating keyways in a horizontal plane.

c. The elevation level vial bubble centered within the thickness of a graduation line.

d. The four quadrant mounting bolts torqued to 25 to 30 foot pounds.

e. The correction counter indicating "00 MILS".

f. The elevation counter indicating "0000 MILS".

3.10 Performance.

3.10.1 Excursion range.

3.10.1.1 Elevation counter. The elevation counter shall cover the excursion of at least 1433 mils in elevation and 280 mils in depression when the correction counter indicates "00 MILS".

3.10.1.2 Correction counter. The correction counter limit stops shall have a minimum of 95 mils excursion, but not over 99 mils excursion on the "+" and "-" dials.

3.11 Accuracy.

3.11.1 Elevation readings. The error at any elevation setting shall not exceed 0.5 mil. The total of all readings shall not exceed 1.0 mil. Nine readings shall be taken in 200 mil increments.

3.11.2 Backlash. Backlash in elevating mechanism shall not exceed 0.7 mil at any elevation setting between 280 mils in depression and 1433 mils in elevation.

3.11.3 Cross level vial bubble. The cross level vial bubble shall remain centered within the thickness of a graduation line when the quadrant is elevated from 280 mils depression to 1433 mils elevation and the elevation level vial bubble is centered.

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3.11.4 Correction counter setting - affect on elevation counter. Setting +50 mils into the correction counter shall decrease the elevation counter reading 50 ± 0.5 mils. Setting -50 mils into the correction counter shall increase the elevation counter reading 50 ± 0.5 mils.

3.11.5 Correction counter setting - affect on elevation level bubble. Any change in position of the elevation level vial bubble shall not be greater than the thickness of a graduation line when the +50 and -50 mil values are set into the correction counter.

3.12 Operability.

3.12.1 Torque. The running torque required to operate the following knobs shall be within the values specified for standard ambient temperature of +60° to +90°F, (+16° to +32°C), and shall not exceed the values specified for extreme operating temperatures:

Knob	At +60°Fto +90°F (+16° to 32°C)	At -50°F and +150°F (-46°C and +65°C)
Elevation	3 to 10 in/lb	13 in/lbs max
Correction	1 to 4 in/lb	8 in/lbs max

3.13 Counter numerals. With the unit digit of each counter centered in the aperture of the cover, the remaining numbers shall be in alignment within one-sixteenth inch.

3.14 Illumination.

3.14.1 Level vial. The level vial bubbles and markings shall be clearly distinguishable when observed in ambient light conditions ranging from dusk into night.

3.14.2 Counters. The numerals on the elevation and correction counters shall be distinguishable when observed under any ambient light conditions, ranging from dusk into night.

3.14.3 External tritium contamination. There shall be no trace of Tritium in excess of 1000 disintegrations per minute (dpm)/100 sq cm on the external surfaces of the quadrant.

3.15 Interchangeability. Quadrant interchangeability shall be verified by insertion, seating, and subsequent removal of the maximum fit gage into the quadrant's mating surface.

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3.16 Reliability. The quadrant shall be subjected to reliability assurance tests as follows:

a. The Quadrant shall meet the requirements of storage and operating temperature of 3.6.1 and 3.6.2.

b. The Quadrant shall be shocked and conform to the requirements of paragraph 3.6.3.

c. The Quadrant shall withstand the Vibration "A" requirements as specified in paragraph 3.6.4.

3.17 Workmanship. The workmanship of MIL-F-13926 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as

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specified in the contract. The sample for first article approval tests shall consist of three (3) quadrants plus three (3) each of all items covered by QAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production. The 3 quadrants shall be tested in accordance with, and meet the requirements of, Tables I and II, and paragraph 4.12.2.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the government, shall inspect the sample to insure that it conforms to all requirements of the contract and submit a record of this inspection, with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the government for approval. The Government reserves the right to witness the contractor's inspection.

4.3. Examination and tests.

4.3.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Quality Assurance Provisions (QAP) listed in the technical data package (TDP). In the absence of QAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to government acceptance or rejection by item or lot.

4.3.3 Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein.

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4.3.4 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

TABLE I. Requirements and test procedures.

No.	Characteristic	Requirement	Test procedure
Critical: None			
Major: 100% Inspection			
101.	Vibration "B"	3.6.5	4.12.4
102.	Cleanliness, defects	3.8.1	4.11
103.	Excursion range, elevation counter	3.10.1.1	4.7.1.2.1
104.	Excursion range, correction counter	3.10.1.2	4.7.1.2.2
105.	Accuracy, elevation readings	3.11.1	4.7.1.3.1
106.	Backlash	3.11.2	4.7.1.3.2
107.	Cross level vial bubble	3.10.3	4.7.1.3.3
108.	Correction counter setting-affect on elevation counter	3.11.4	4.7.1.3.4
109.	Correction counter setting-affect on elevation level bubble	3.11.5	4.7.1.3.5
110.	Operability, torque (+60° to +90°)	3.12.1	4.8.1
111.	Counter numerals	3.13	4.9
112.	Illumination, level vial	3.14.1	4.10.1
113.	Illumination, counters	3.14.2	4.10.2
114.	Interchangeability	3.15	4.13
115.	Sealing	3.7.1	4.14.1
116.	Purging	3.7.2	4.14.2

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No.	Characteristic	Requirement	Test procedure
117.	External tritium contamination	3.14.3	4.10.3
118.	Fabrication	3.4	11729525
119.	Packaging	5.1	4.15

Minor: None

4.4 Special sampling.

4.4.1 General. Subsequent to meeting the requirements of Table I, three quadrants shall be selected at random by a Government representative as a special sample from each 50 produced or from each month's production, whichever occurs first. The samples shall meet the requirements and tests in Table II and shall meet the requirements and tests in Table I, except characteristic 101, after being subjected to Table II testing.

TABLE II. Requirements and test procedures.

No.	Characteristic	Requirement	Test procedure
301.	Storage temperature	3.6.1	4.12.1
302.	Operating temperature	3.6.2	4.12.1
303.	Torque (+150°F & -50°F)	3.12.1	4.8.1

4.4.2 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the government until necessary correction have been made by the contractor and the resubmitted sampling have been approved (See 4.3.3).

4.5 Test equipment.

4.5.1 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

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4.5.3 Contractor furnished inspection equipment.

4.5.3.1 Government design. Unless otherwise stated in the contract, all inspection equipment specified by drawing number in specifications or QAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.3.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3.3 Test equipment. In conjunction with 4.5.3.2, the following standard test equipment shall be utilized in the performance of the applicable test as specified in 4.8.

Nomenclature	Description
1. Hot and cold chamber	Standard type conforming to the accuracies outlined under in MIL-F-13926.
2. Torque wrench	Standard shop type accurate to 10% of the drawing tolerance.

4.6 Reliability assurance sample. Unless otherwise specified in the contract, the Government shall be responsible for reliability assurance testing. The reliability assurance test sample shall consist of one quadrant selected at random by the Government from any accepted regular production lot prior to delivery of the first 25% of the basic contract quantity. All quadrants previously selected for First Article Test or Special Sampling shall be excluded from the lot for purposes of selecting the reliability assurance test sample.

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4.6.1 Reliability assurance testing. After random selection of the sample, the contractor shall provide to the Government the inspection records, including certificates of conformance, for the sample. The contractor shall also provide a list of all changes, deviations, and waivers under the contract. The sample shall be tested for reliability as represented by the requirements of 3.6.1 through 3.6.4. Testing shall be performed in accordance with the procedures specified in 4.12.1 through 4.12.3.

4.6.2 Defects. Defects resulting from reliability assurance testing may be assessed as sufficient evidence that the contractor's production processes and quality control procedures do not provide adequate assurance against product reliability degradation. Upon identification by the government to the contractor, of the specific quality defect(s) found, the contractor shall determine and apply effective corrective action to improve his production processes and quality control procedures as necessary to eliminate the defect(s) from quadrants not yet delivered under the contract.

4.6.3 Retest. Whenever reliability assurance test results require corrective action by the contractor, a retest sample shall be selected in accordance with 4.6 except that the sample shall be selected from the first 10 quadrants produced subsequent to implementation of the corrective action. The retest sample shall be subjected only to the reliability test(s) which revealed the defect(s).

4.6.3.1 Inspection after retest. Inspection of the sample after retest shall be conducted by the Government and shall be in accordance with 4.3.2.2, "Final Acceptance Inspection", except characteristics 101 of Table I, 301 and 302 of Table II. Reappearance of the previous defect(s), 4.6.2, may be cause for the Government to prescribe mandatory process corrective action by the contractor at no additional cost to the Government.

4.7 Methods of inspection.

4.7.1 Performance.

4.7.1.1 Orientation. Final inspection fixture 11747194 and inspection aid 8658940 shall be used for inspection of the quadrant. The inspection fixture shall be positioned on a vibration free surface and set-up in accordance with the set-up instructions contained on the inspection fixture drawing. Subsequent to completion of all set-up and mounting instructions, the quadrant shall be oriented as specified in 3.9 of this specification.

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4.7.1.2 Excursion range.

4.7.1.2.1 Elevation counter. With the elevation counter indicating "000 MILS", the elevation counter knob shall be manually rotated until the full excursion has traveled the minimum elevation and depression specified in 3.10.1.1, as read on the elevation counter.

4.7.1.2.2 Correction counter. With the correction counter indicating "00 MILS", the correction counter knob shall be manually rotated until the full excursion has traveled the minimum and maximum limits specified in 3.10.1.2, as read on the "+" and "-" dials of the correction counter.

4.7.1.3 Accuracy.

4.7.1.3.1 Elevation readings. Set the correction counter to indicate "00 MILS" and the elevation counter to indicate "200 MILS" depression. With backlash removed for direction of travel, rotate elevation fixture 11729722 until the elevation level vial bubble of the quadrant under test is centered within the thickness of a graduation line. Do not overtravel the center position. Place inspection aid 8658940 on Surface "A" shown on Drawing F11747194. Measure and record the angular setting in the elevation fixture. Repeat the above procedure with the elevation counter set to indicate "0000 MILS", "200 MILS", "400 MILS", "600 MILS", "800 MILS", "1000 MILS", "1200 MILS", and "1400 MILS" elevation when the correction counter indicates "00 MILS". The recorded readings shall indicate the elevation counter settings within the values specified in 3.11.1, as measured on the inspection aid.

4.7.1.3.2 Backlash. With the quadrant mounted and oriented as specified in 3.9, set the elevation counter to "0000" by turning the elevation knob clockwise. Do not over travel the "0000" setting. Adjust fixture elevation to center quadrant level vial bubble within the requirement of 3.9c. Measure and record angular setting with gunner quadrant on the gunner quadrant seat of fixture, 11747194. Rotate the elevation knob a minimum of one-half turn clockwise and reset the elevation counter to "0000" by turning knob counter clockwise. Do not overtravel the "0000" setting. Adjust elevation fixture to center the quadrant level vial bubble within the requirements of 3.9c. Measure and record the angular setting the same as above. The backlash is the difference between the two readings, and shall meet the requirements of 3.11.2. In addition, the above test shall be performed at -200, +200, 400, 600, 800, 1000, 1200 and 1400 mils, and can be performed in conjunction with 4.7.1.3.1. (There shall be no undue pressure exerted on the housing or the worm shaft, up or down, as the wormshaft is being turned to perform the above tests.)

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4.7.1.3.3 Cross level vial bubble. The position of the cross level vial bubble shall conform to the requirements of 3.11.1 when observed during the test specified in 4.7.1.3.1.

4.7.1.3.4 Correction counter setting, affect on elevation counter. This test shall be performed utilizing the method indicated in 4.7.1.3.1. Position the quadrant as specified in 4.7.1.1. Rotate the correction counter to +50 mils setting. The elevation counter reading shall decrease an equal amount within the limits specified in 3.11.4, as measured on the inspection aid. Rotate the correction counter to -50 mils setting. The elevation counter reading shall increase an equal amount within the limits specified in 3.11.4, as measured on the inspection aid.

4.7.1.3.5 Correction counter setting, affect on elevation level bubble. The position of the elevation level bubble shall conform to the requirements of 3.11.5 when observed during the test specified in 4.7.1.3.4.

4.8 Operability.

4.8.1 Torque. The running torque required to rotate the knobs specified in 3.12.1, at standard ambient and extreme operating temperatures, shall be determined with a standard torque measuring device. The torque measuring device shall be equipped with the following adapters to fit the respective knobs being tested. The measured torque values shall be within the limits specified in 3.12.1.

Knob	Adapter number
Correction Counter	B11747940
Elevation	B11747939

4.9 Counter numerals. All readings shall be centered and a standard measuring scale shall be used to determine numeral (digit) alignment within the linear value specified in 3.13.

4.10 Illumination.

4.10.1 Level vial. The level vial illumination test shall be performed under simulated conditions specified in 3.14.1. Place the instrument in a dark room having a controlled light source. The observer and instrument must become dark adapted. When this has been accomplished, the unaided eye shall be capable of distinguishing the level vial bubble and markings at various light levels that simulate dusk into darkness.

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4.10.2 Counters. Perform this test in conjunction with the test in 4.10.1. The unaided eye shall be capable of determining the counter numerals and numeral alignment at various light levels that simulate the light conditions of 3.14.2.

4.10.3 External tritium contamination. Contamination test shall be performed by wiping the quadrant with filter paper moistened in distilled water. The wiping shall be performed with moderate finger pressure. The damp filter paper shall be placed in a container with the proper portion of scintillation liquid. The scintillating system used to measure contamination must be calibrated. The read out shall be within 10% of the known standard value when counted to a total of 2000 disintegrations. The actual test for determining contamination is performed by placing the container into the scintillation system. The measured contamination shall be within the limits specified in 3.14.3.

4.11 Cleanliness. The quadrant under test shall be subjected to a visual examination to determine conformance to the requirements outlined in 3.8.

4.12 Environmental.

4.12.1 Storage and operating temperatures. The quadrant shall be subjected to one cycle of temperature variations in accordance with Table III. Upon completion of sequence 1 and 4 soak periods, (storage), the quadrant shall be visually and tactually examined to insure conformance with requirements of 3.6.1. Upon completion of sequence 2 and 4 soak periods, (operating), the quadrant shall be visually and tactually examined to assure conformance with the requirements of 3.12.1. Following completion of sequence 3 and 5 soak periods, (room temperature), the quadrant shall meet all the requirements of 3.7 through 3.15.

TABLE III. Storage and operating temperatures

SEQ	STORAGE TEMP	SOAK	OPERATING TEMP	SOAK	ROOM TEMP	SOAK
1	+160°F (+71°C)	4 hrs				
2			+150°F (+65°C)	2 hrs		
3					+60°F to +90°F (+16°C to 32°C)	4 hrs
4	-50°F (-46°C)	4 hrs	-50°F (-46°C)	4 hrs*		
5					+60°F to +90°F (+16°C to 32°C)	4 hrs

*Same 4 hrs as storage soak

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4.12.2 Shock. This test is applicable to the First Article Samples and the Reliability Assurance Samples only. The quadrant shall be positioned on Fixture 11747195 and secured to the shock tester. The quadrant shall be subjected to a shock test in accordance with 3.6.3. Upon completion of the shock, the quadrant shall be examined and inspected to the requirements specified in 3.6.3.

4.12.3 Vibration "A". This test is applicable to the Reliability Assurance Sample only. The quadrant shall be positioned on Fixture 11747195 and secured to the vibration tester. The quadrant shall be subjected to a vibration test in accordance with 3.6.4. Upon completion of the vibration test, the quadrant shall be examined and inspected to the requirements of 3.6.4.

4.12.4 Vibration "B". The quadrant shall be positioned on Fixture 11747195 and vibrated in accordance with the frequency and duration as specified in 3.6.5. At the conclusion of the test, the quadrant shall be subjected to a visual and tactile examination and shall meet the requirements of 3.6.5.

4.13 Interchangeability. The interchangeability gage 11747969 shall be applied to the mounting surface (keyway) of the quadrant to assure maximum fit condition. The gage shall be firmly seated in the keyway, without binding, and subsequently removed, to meet the requirements of 3.15.

4.14 Sealing and purging.

4.14.1 Sealing. The quadrant shall be capable of withstanding an internal pressure of 5 + 0.10 PSIG for a period of 1 hour with no evidence of leakage. Prior to the start of the test, secure the poppet valve in a closed position. Then pressurize the quadrant with dry nitrogen as specified in 3.7.1. Accuracy of the test equipment used for this test shall be equal to or exceed the accuracies depicted on Pressure Test Drawing F8565556.

4.14.2 Purging. Connect the test equipment to the purging valve of the quadrant. Adjust the dry nitrogen pressure to the pressure specified in 3.7.2. The pressure is to remain constant (8.0 $\pm$ 1/4 psi) until the dewpoint of the emergent gas is no higher than -25°F. Disconnect the purging apparatus and maintain the internal pressure required for testing and delivery to meet the requirements of 3.7.2.

4.15 Packaging inspection. The preservation, packing, and marking shall be inspected to verify conformance to the requirements of 5.1.

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5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 11729525. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Quadrant, Fire Control M18 when attached to a telescope and quadrant mount is used for laying the artillery weapon in elevation for indirect fire.

6.2 Acquisition requirements. Procurement data should specify the following:

a. Title, number and date of this specification.

b. Selection of an applicable level of preservation, packaging, and packing.

c. Applicable packaging data sheet number (See 5.1).

d. Applicable stock number.

e. Provisions for First Article Testing.

f. Reliability Assurance Sample.

g. Contract data requirements for submission of inspection equipment designs conforming to Data Item Description DI-R-1714. See 6.3.1.

h. Prospective bidders should be made aware that a Nuclear Regulatory Commission License is required for the manufacture of this item.

i. Levels of preservation and packaging. (See 5.1).

6.3 Inspection equipment design.

6.3.1 Submission of designs for approval. Contractor designs for final acceptance inspection shall be approved by the Government prior to fabrication or procuring the equipment. The contractor is referred to MIL-HDBK-204 for guidance. Submission of design concept on inspection equipment is permissable for tentative approval. The completion date for design review will be based on the data of the

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final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply. When the contractor submits inspection equipment designs to the government for approval, he shall give the following information in his letter of transmittal:

a. The contract number.

b. The contract item (name, model number, etc.).

c. The designs remaining to be submitted and the expected date of submittal.

6.4 Drawings. Drawings listed in section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Conrol
Quadrant
Radioluminous

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Certification statement. This specification is certified to be in compliance with current Army Material Command (AMC) policy for the elimination of AQLs/LTDPs (Acceptance Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

(Certified by/date: _____.)

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1290-A430)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-Q-48556A (AR)	26 November 1989

3. DOCUMENT TITLE
QUADRANT, FIRE CONTROL: M18

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US Army ARDEC Standardization & Specification Office	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S, Bldg 6 Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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